# Waves

30 questions: 10 multiple choice, 20 structured.

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## 1 · Part 1

Paper 1 & 2 Units [1]

Which row shows the correct unit for each quantity?

| | frequency | wave speed | time period |
|---|---|---|---|
| A | Hz | m/s | s |
| B | s | m/s | Hz |
| C | Hz | m | s |
| D | m/s | Hz | s |

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## 2 · Part 1

Paper 1 & 2 Properties of waves [1]

What is the wavelength of a wave?

A the distance from a crest to the next trough

B the distance between a point on one wave and the same point on the next wave

C the vertical distance from the top of a crest to the bottom of a trough

D the distance that the wave travels in one second

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## 3 · Part 1 Paper 1 & 2 Properties of waves [1]

Radio waves travel at $3.0 \times 10^8$ m/s. A radio transmitter emits waves with a frequency of $1.2 \times 10^8$ Hz.

What is the wavelength of these radio waves?

**A** $3.6 \times 10^{16}$ m

**B** 25 m

**C** 2.5 m

**D** 0.40 m

## 4 · Part 1 Paper 1 & 2 Properties of waves [1]

The whistle on a train emits sound of a constant frequency. The train moves towards a person standing still on a platform.

Which row describes the sound that reaches the person, compared with the sound emitted by the whistle?

| | **frequency** | **wavelength** |
|---|---|---|
| **A** | higher | shorter |
| **B** | higher | longer |
| **C** | lower | shorter |
| **D** | lower | longer |

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## 5 · Part 1 Paper 1 & 2 The electromagnetic spectrum [1]

Which list shows colours of visible light in order of increasing frequency?

**A** violet, indigo, blue, green

**B** red, yellow, orange, green

**C** orange, yellow, green, blue

**D** blue, green, yellow, orange

## 6 · Part 1

Paper 1 & 2 | The electromagnetic spectrum | [1]

Which row correctly gives a use of the radiation and a harmful effect of excessive exposure to it?

| | radiation | use | harmful effect |
|---|---|---|---|
| A | microwaves | sterilising food | skin burns |
| B | infrared | night vision equipment | skin burns |
| C | ultraviolet | satellite transmissions | blindness |
| D | gamma rays | fluorescent lamps | cancer |

## 7 · Part 1

Paper 1 & 2 | Light and sound | [1]

Light travels from air into glass. The refractive index of the glass is 1.5. The angle of refraction in the glass is 30°.

What is the angle of incidence in the air?

A 0.75°

B 19°

C 45°

D 49°

## 8 · Part 1

Paper 1 & 2 | Light and sound | [1]

Material X has a larger refractive index than material Y.

Which statement about the critical angles of X and Y is correct?

A The critical angle of X is larger than the critical angle of Y.

B The critical angle of X is smaller than the critical angle of Y.

C The critical angles of X and Y are the same.

D Only material Y has a critical angle.

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## 9 · Part 1 Paper 2 only Light and sound [1]

An animal can hear sounds with frequencies from 60 Hz to 45 000 Hz.

Which frequency of sound could be heard by this animal but not by a human with normal hearing?

A 30 000 Hz

B 10 000 Hz

C 40 Hz

D 50 000 Hz

## 10 · Part 1 Paper 2 only Light and sound [1]

One end of a ruler is clamped to the edge of a bench and the free end is made to vibrate, producing a sound. The ruler is then adjusted so that it vibrates with a higher frequency and a smaller amplitude.

Which row describes the new sound, compared with the first sound?

| | **loudness** | **pitch** |
|---|---|---|
| A | louder | higher |
| B | louder | lower |
| C | quieter | lower |
| D | quieter | higher |

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## 11 · Part 1

Paper 1 & 2 | Properties of waves | [5]

Straight water waves are produced in a ripple tank by a wooden bar that vibrates up and down on the surface of the water. One small cork floats on the water. As the waves pass, the cork moves up and down but does not move along the tank.

(a) State whether water waves are transverse or longitudinal. [1]

(b) Explain what the movement of the cork shows about what waves transfer. [2]

(c) State what is meant by the amplitude of a wave. [1]

(d) State what is meant by a wavefront. [1]

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## 12 · Part 1

Paper 1 & 2 | Properties of waves | [6]

Sara holds one end of a long stretched spring that lies on a smooth bench. The other end of the spring is fixed to a wall.

(a) Describe how Sara should move her hand to produce a longitudinal wave along the spring. [1]

(b) Explain the difference between a longitudinal wave and a transverse wave. [2]

(c) Sara moves her hand backwards and forwards 4 times every second. The distance between the centres of neighbouring compressions on the spring is 0.30 m.

Calculate the speed of the wave along the spring. [3]

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## 13 · Part 1

Paper 1 & 2

Properties of waves

[8]

Hassan strikes a tuning fork. It produces a sound wave in air with a time period of 0.0025 s.

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(a) State what is meant by the frequency of a wave. [1]

(b) Calculate the frequency of the sound wave.

Use the relationship: frequency = 1 / time period [2]

(c) The speed of sound in air is 340 m/s.

Calculate the wavelength of the sound wave in air. [3]

(d) Hassan then strikes a different tuning fork that vibrates with a higher frequency.

Explain what happens to the wavelength of the sound in air. [2]

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## 14 · Part 1

Paper 1 & 2 Properties of waves [8]

One buoy floats in the sea. Waves pass the buoy and make it move up and down. The buoy makes 12 complete up-and-down oscillations in 60 s. The distance between neighbouring wave crests is 15 m.

(a) Show that the frequency of the waves is 0.20 Hz.

Use the relationship: frequency = 1 / time period [2]

(b) Calculate the speed of the waves. [3]

(c) Closer to the beach the water is shallower. The waves travel at 2.4 m/s in the shallow water. Their frequency does not change.

Calculate the wavelength of the waves in the shallow water. [2]

(d) When the waves pass at an angle from deep water into shallow water, they change direction.

State the name of this effect. [1]

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## 15 · Part 1

Paper 1 & 2 Properties of waves [6]

Lena stands still at the side of a straight road. A car drives along the road at constant speed, towards Lena and then past her. The driver sounds the horn the whole time. The horn emits sound of a constant frequency.

(a) State the name of the effect that causes the frequency heard by Lena to be different from the frequency emitted by the horn. [1]

(b) Describe and explain how the frequency of the sound heard by Lena compares with the frequency emitted by the horn, while the car is approaching her and after it has passed her. [5]

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## 16 · Part 1

Paper 1 & 2 | Properties of waves | [6]

Ships can measure the depth of water by sending pulses of sound down to the seabed. One ship sends a pulse of sound vertically downwards. The pulse returns to the ship 0.80 s after it was sent. The speed of sound in seawater is 1500 m/s.

(a) State what happens to the pulse of sound at the seabed. [1]

(b) Calculate the depth of the water below the ship. [3]

(c) When sound passes from the seawater into the rock of the seabed, some of the sound is refracted.

State what is meant by refraction of a wave. [2]

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## 17 · Part 1

Paper 1 & 2 | The electromagnetic spectrum | [6]

The table lists the regions of the electromagnetic spectrum in order of decreasing wavelength. Two regions are missing and are labelled X and Y.

| position | region |
|---|---|
| 1 | radio waves |
| 2 | microwaves |
| 3 | X |
| 4 | visible light |
| 5 | ultraviolet |
| 6 | Y |
| 7 | gamma rays |

(a) State the names of regions X and Y. [2]

(b) State which region in the table has the highest frequency. [1]

(c) State how the speed of microwaves in free space compares with the speed of gamma rays in free space. [1]

(d) Describe one use of radio waves and one use of ultraviolet. [2]

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## 18 · Part 1 Paper 1 & 2 The electromagnetic spectrum [8]

Electromagnetic radiations have many uses, but excessive exposure to some of them is harmful.

(a) X-rays are used in hospitals to observe the internal structure of the body.

Explain why X-rays can be used to produce an image of the bones inside the body. [2]

(b) Gamma rays are used to sterilise medical equipment that has already been sealed in plastic packaging.

Explain why gamma rays are suitable for this use. [2]

(c) Jamal works outdoors in strong sunlight for many hours each day.

Describe the harmful effects of excessive exposure to ultraviolet radiation, and simple measures Jamal could take to protect himself. [4]

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## 19 · Part 1

Paper 1 & 2 | Light and sound | [5]

Priya shines a ray of light at a plane mirror. The angle between the incident ray and the surface of the mirror is 35°.

(a) State the law of reflection. [1]

(b) Determine the angle of incidence. [1]

(c) Determine the angle between the incident ray and the reflected ray. [2]

(d) State whether light waves are transverse or longitudinal. [1]

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## 20 · Part 1

Paper 1 & 2 Light and sound [7]

Mei investigates the refraction of light using a rectangular glass block, a ray box and a protractor. She places the block on a sheet of paper and directs a ray of light into one face of the block. She marks the paths of the rays on the paper, draws the normal where the ray enters the block, and measures the angle of incidence i and the angle of refraction r.

Her results are shown in the table.

| i / ° | r / ° |
|---|---|
| 20 | 13 |
| 40 | 25 |
| 60 | 35 |

(a) State the independent variable in this investigation. [1]

(b) Describe how Mei can find the path of the ray inside the block, so that she can measure the angle of refraction. [2]

(c) Calculate the refractive index of the glass, using her result for i = 40°. [3]

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(d) Suggest why Mei took readings for several different angles of incidence. [1]

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## 21 · Part 1 Paper 1 & 2 Light and sound [6]

Tomas uses a semicircular glass block to investigate what happens to light at a boundary between glass and air. He directs a ray of light through the curved surface of the block, towards the centre of the flat face. The ray passes into the block without changing direction. He changes the angle of incidence at the flat face and records his observations.

| **angle of incidence at flat face / °** | **observation** |
|---|---|
| 20 | a bright ray leaves the flat face into the air; a faint ray is reflected inside the block |
| 30 | a bright ray leaves the flat face into the air; a faint ray is reflected inside the block |
| 41 | the ray leaving the block travels along the flat face; a ray is reflected inside the block |
| 50 | no ray leaves the flat face; a bright ray is reflected inside the block |

(a) Give a reason why the ray does not change direction when it enters the curved surface of the block. [1]

(b) Explain the meaning of the critical angle. [2]

(c) State the critical angle of this glass, using Tomas's observations. [1]

(d) Calculate the refractive index of the glass. [2]

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## 22 · Part 1 Paper 1 & 2 Light and sound [8]

The refractive index of water is 1.33.

(a) One ray of light in air meets the surface of some water with an angle of incidence of 50°.

Calculate the angle of refraction in the water. [3]

(b) Explain why the ray changes direction as it enters the water. [2]

(c) Under the water, a diver shines a torch upwards. The light meets the water surface from below, with an angle of incidence of 55°.

Determine whether any of this light passes out of the water into the air. [3]

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## 23 · Part 1

Paper 1 & 2 Light and sound [6]

Optical fibres are thin strands of glass. They are used to carry information, such as internet data, as pulses of light.

(a) Describe the role of total internal reflection in transmitting information along an optical fibre. [4]

(b) Some periscopes use glass prisms instead of mirrors. Inside one prism, light meets a glass–air surface with an angle of incidence of 45°. The critical angle of the glass is 42°.

Explain why all of the light is reflected at this surface. [2]

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## 24 · Part 1 Paper 1 & 2 Light and sound [5]

Sound waves travel through air.

(a) State whether sound waves are transverse or longitudinal. [1]

(b) Describe how the air particles move as a sound wave passes through the air. [2]

(c) On a still evening, the air near the ground can be colder than the air higher up. Sound travels more slowly in cold air than in warm air.

Explain why sound waves can change direction as they travel from the colder air into the warmer air. [2]

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## 25 · Part 1

Paper 2 only

Light and sound

[7]

Ali and Beth measure the speed of sound in air. They stand 200 m apart on a playing field. Ali bangs two wooden blocks together. Beth starts a stopwatch when she sees the blocks hit each other and stops it when she hears the sound. They do this five times.

| trial | 1 | 2 | 3 | 4 | 5 |
|---|---|---|---|---|---|
| time / s | 0.62 | 0.58 | 0.60 | 0.64 | 0.56 |

(a) Calculate the mean time. [1]

(b) Calculate the speed of sound in air from these results. [3]

(c) Explain why it is acceptable for Beth to start the stopwatch when she sees the blocks hit each other. [2]

(d) Suggest one way the students could make their value for the speed of sound more accurate. [1]

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## 26 · Part 1

Paper 2 only

Light and sound

[6]

One loudspeaker is connected to a signal generator and produces a sound of constant frequency. Kofi places a microphone in front of the loudspeaker and connects the microphone to an oscilloscope, so that the sound wave is displayed as a trace on the screen.

(a) Describe what the microphone does. [2]

(b) State the quantity shown on the vertical axis of the oscilloscope screen and the quantity shown on the horizontal axis. [2]

(c) Kofi makes the sound louder without changing its frequency.

Describe how the trace on the screen changes. [2]

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## 27 · Part 1

Paper 2 only

Light and sound

[7]

Zara uses a microphone connected to an oscilloscope to measure the frequency of the sound from a whistle. The time-base of the oscilloscope is set so that each horizontal division represents 0.0005 s. One complete wave on the trace occupies 4.0 horizontal divisions.

(a) Calculate the time period of the sound wave. [2]

(b) Calculate the frequency of the sound wave.

Use the relationship: frequency = 1 / time period [2]

(c) Zara changes the time-base so that each horizontal division represents 0.00001 s. A different source produces a trace on which one complete wave occupies 2.5 horizontal divisions.

Determine whether a person with normal hearing could hear the sound from this source. [3]

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## 28 · Part 1

Paper 2 only

Light and sound

[6]

The table gives information about the vibrations of two strings, X and Y, on a musical instrument when they are plucked.

| string | frequency of vibration / Hz | amplitude of vibration |
|---|---|---|
| X | 220 | large |
| Y | 660 | small |

(a) State which string produces the higher-pitched sound. Give a reason for your answer. [2]

(b) State which string produces the louder sound. Give a reason for your answer. [2]

(c) After string X is plucked, its sound gradually becomes quieter but its pitch does not change.

Explain these observations in terms of the vibration of the string. [2]

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## 29 · Part 1

Paper 2 only

Light and sound

[5]

Nadia connects a loudspeaker to a signal generator. She slowly increases the frequency of the signal from 10 Hz to 30 000 Hz. Nadia has normal hearing.

(a) State the frequency range of normal human hearing. [2]

(b) Describe what Nadia hears as the frequency increases from 10 Hz to 30 000 Hz. [3]

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## 30 · Part 1

Paper 2 only

Light and sound

[6]

Leo and Maya measure the speed of sound in air using echoes. Leo stands 85 m from a tall flat wall and claps his hands. He adjusts the rate of clapping until each clap happens at the same moment as he hears the echo of the previous clap. The time between two claps is then equal to the time for the sound to travel to the wall and back.

Maya uses a stopwatch to measure the time for 20 intervals between claps. Her reading is 10.0 s.

(a) Calculate the time for one interval between claps. [1]

(b) Calculate the speed of sound in air. [3]

(c) Explain why Maya timed 20 intervals instead of timing one interval. [2]

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